Task 1 : SQL – JOIN and Aggregation

- Create the Employees and Departments tables
- Insert 8+ employees and 4 departments (one with no employees)
- Show department name, employee count, and highest salary
- Use LEFT JOIN and GROUP BY

```sql
CREATE DATABASE Company;
USE Company;
CREATE TABLE Employees (
emp_id INT PRIMARY KEY,
emp_name varchar(200),
department_id INT,
Salary DECIMAL );

CREATE TABLE Departments(
department_id INT PRIMARY KEY,
department_name varchar(200));

INSERT INTO Employees VALUES
(101,'Arun',1,40000),
(102,'Bala',2,60000),
(103,'Chitra',2,75000),
(104,'Dev',3,50000),
(105,'Esha',1,45000),
(106,'Farid',2,65000),
(107,'Gita',3,55000),
(108,'Hari',1,42000);

INSERT INTO Departments VALUES
(1,'HR'),
(2,'IT'),
(3,'Finance'),
(4,'Marketing');
```

```sql
(1,'HR'),
(2,'IT'),
(3,'Finance'),
(4,'Marketing');


SELECT d.department_name, COUNT(e.emp_id) AS num_employees, MAX(e.salary) AS highest_salary
FROM departments d
LEFT JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.department_name;

USE Company;
```

department_name	num_employees	highest_salary
HR	3	45000
IT	3	75000
Finance	2	55000
Marketing	0	NULL

Task 2: Linux – Searching Text with grep

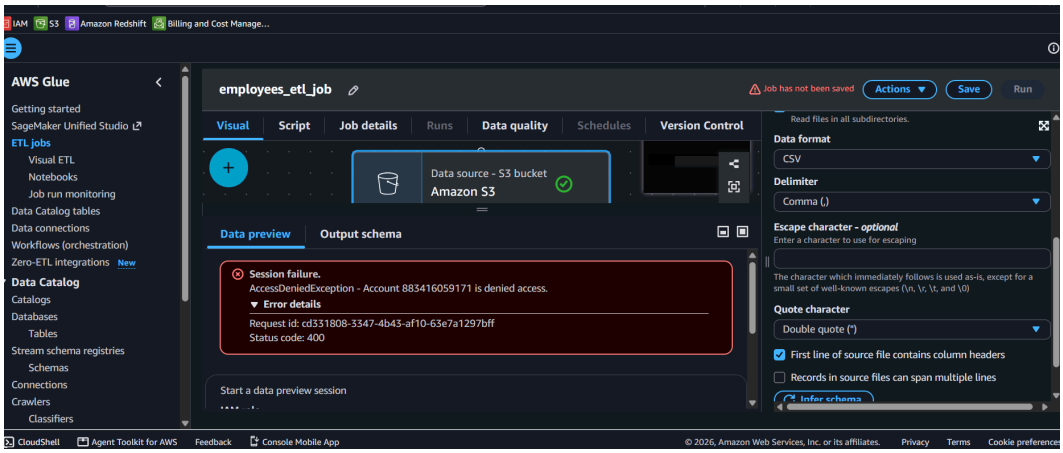
- Create data.txt with 5+ lines
- Search a word using grep
- Show the matching lines

```
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$ cat data.txt
cat: data.txt: No such file or directory
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$ touch data.txt
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$ cat > data.txt << 'EOF'
Linux is an open source operating system
AWS provides cloud services
Glue is used for ETL in AWS
Python is popular for data engineering
Azure offers Data Lake Storage
EOF
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$ cat data.txt
Linux is an open source operating system
AWS provides cloud services
Glue is used for ETL in AWS
Python is popular for data engineering
Azure offers Data Lake Storage
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$ grep "AWS" data.txt
AWS provides cloud services
Glue is used for ETL in AWS
darys@LAPTOP-6VDOQ70I:/mnt/c/Users/ASUS$
```

Task 3: AWS Glue ETL – Data Transformation

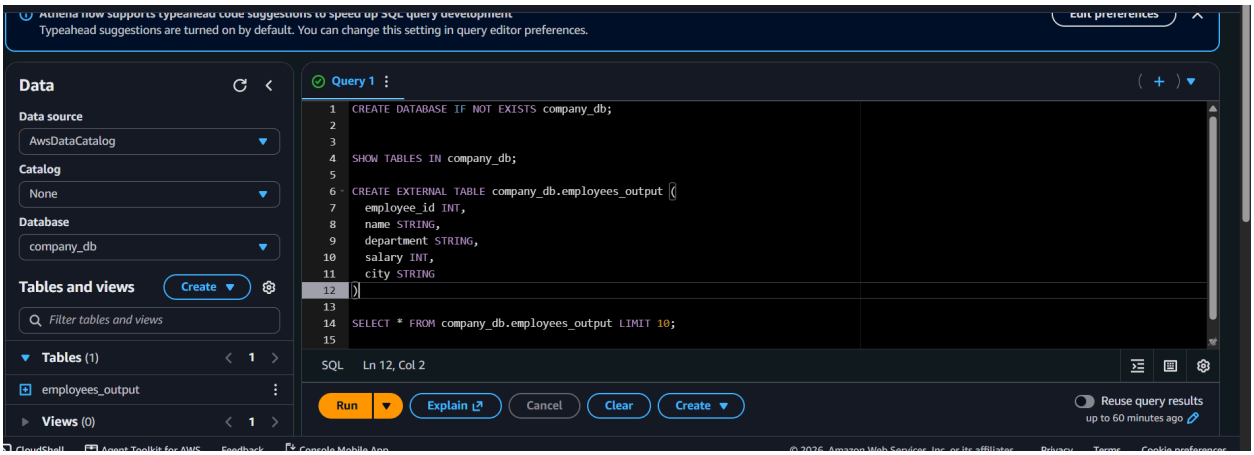
- Using an existing AWS Glue job and a prepared dataset, identify the S3 source.
- Run an ETL job that reads the dataset and performs a simple transformation, such as selecting columns or filtering rows.
- Write the transformed dataset to another S3 location.
- Expected output: the transformed dataset stored in the destination location.

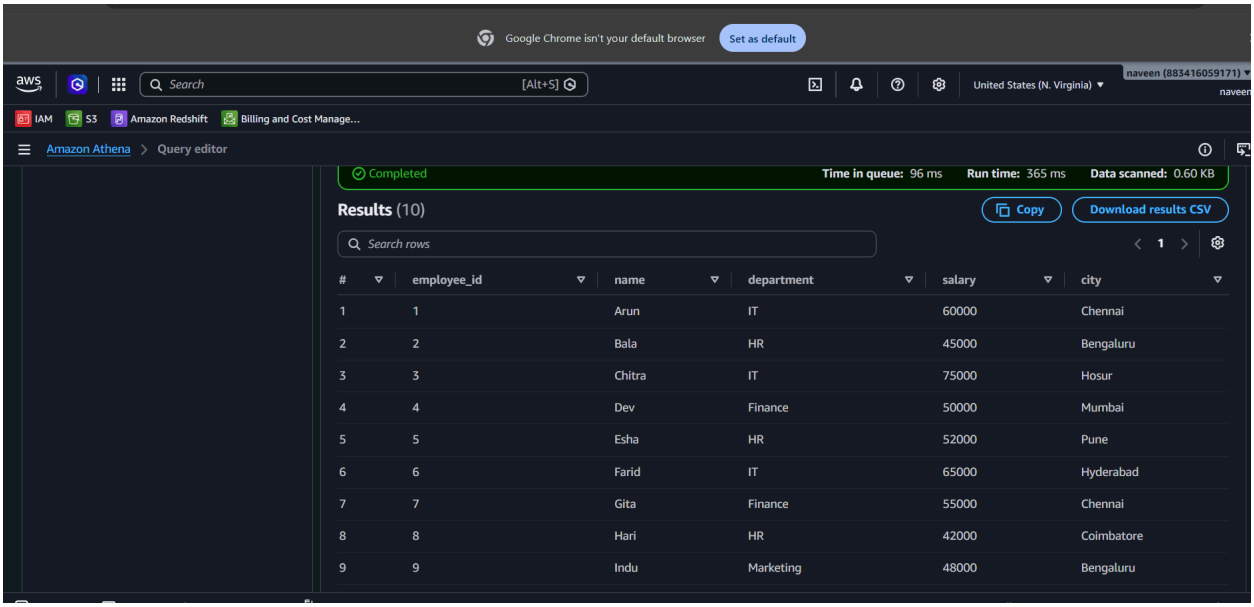
- I am getting access denaid for the Glue ETL task



Task 4: Amazon Athena – SQL Query Execution

- Open Amazon Athena and select an available database and a table containing data stored in S3.
- Execute a SELECT query and demonstrate the results.
- Expected output: the records returned by the query.





Task 5: Azure Data Lake Storage Gen2

- Open an Azure Storage Account with hierarchical namespace enabled.
- Create or select a filesystem and upload a text file.
- Expected output: the file listed in the filesystem.

